

MA 546: Introduction to Quantitative Finance

Course Code: MA 546

Course Name: Introduction to Quantitative Finance

L-T-P-C: 3-1-0-4

Prerequisites: Probability and Statistics, Linear Algebra, Optimizations, Programming Language

Intended for: B.Tech/ MS/PhD

Mutual Exclusion: NA

Elective or Core: Elective

Approval: 60th BoA

Preamble

This course introduces students to the quantitative foundations of investment and portfolio management. It integrates concepts from mathematics, statistics, and optimization with real-world financial applications, enabling students to understand and model the behavior of financial markets. The course begins with an overview of financial instruments and markets, followed by mathematical preliminaries essential for investment analysis. It then delves into core topics such as portfolio optimization, risk measures, pricing models, and option valuation. Emphasis is placed on the quantitative modeling of investments, including mean-variance analysis, Capital Asset Pricing Model (CAPM), and the introduction to Black-Scholes framework. By the end of the course, students will be equipped to analyze, construct, and evaluate investment portfolios with a strong quantitative grounding. This elective is ideal for students interested in careers or research in quantitative finance, financial engineering, or investment analytics.

Course Modules

- **Financial Markets and Instruments:** Introduction to Financial Markets, Stock Exchanges and IPOs: Functions of exchanges, IPO process. Market Indices calculation and interpretation. Time Value of Money, Financial Instruments: Stocks, Bonds, and Derivatives. (6 Lectures)
- **Mathematical Preliminaries:** A quick review of essential concepts from Linear Algebra, Probability and Statistics, and convex Optimization. (4 Lectures)
- **Portfolio Optimization Foundations:** Asset returns and properties, Risk types and key measures, Two-asset and multi-asset portfolio, Optimal portfolio return and efficient frontier, Portfolio with a risk-free asset, tangency portfolio, Capital allocation line (CAL). (8 Lectures)
- **Risk Measures:** Types of Risk and Risk Measures, Volatility, Value at Risk (VaR): properties and limitations. Conditional VaR (CVaR): Definition, calculation, and advantages over VaR, Skewness and Kurtosis, Systematic risk measure: calculation, and interpretation. (4 Lectures)
- **Optimization, Pricing Models, and Performance:** Capital Asset Pricing Model (CAPM): Assumptions, beta, Security Market Line, introduction to factor models. Portfolio Performance Metrics, Mean-Variance Optimization: the mathematical framework, Portfolio with Practical Constraints: Short-selling constraints, cardinality constraints, and transaction costs. Portfolio optimization with Factor Models, Black-Litterman Model, Portfolio Rebalancing Strategies. (12 Lectures)
- **Option Pricing Fundamentals:** Arbitrage & Risk-Neutral Valuation, Introduction to Black-Scholes Model, European vs. American Options, Binomial Tree Model and convergence to Black-Scholes, In-



roduction to stochastic processes, geometric Brownian motion as a stock price model, intuitive link to Black–Scholes. (8 Lectures)

Laboratory/Practical/Tutorial Modules

This course will also include one tutorial session every week dedicated to solving exercises relevant to the studied topics.

Text Books

1. Elton, E. J., Gruber, M. J., Brown, S. J., and Goetzmann, W. N. *Modern Portfolio Theory and Investment Analysis*. Wiley, 9th Edition, 2014.
2. Luenberger, D. G. *Investment Science*. Oxford University Press, 2nd Edition, 2014.

References

1. Hull, J. C. *Options, Futures, and Other Derivatives*. Pearson Education, 11th Edition, 2022.
2. Bodie, Z., Kane, A., and Marcus, A. J. *Investments*. McGraw Hill, 13th Edition, 2024.